

§11.3 - The Integral Test and Estimates of Sums

Theorem #1

Let f be a positive, decreasing, continuous function on $[1, \infty)$ and define (a_n) by

$$a_n = f(n).$$

Then the following are equivalent:

1. The improper integral

$$\int_1^{\infty} f(x) \, dx$$

converges

2. The series

$$\sum a_n$$

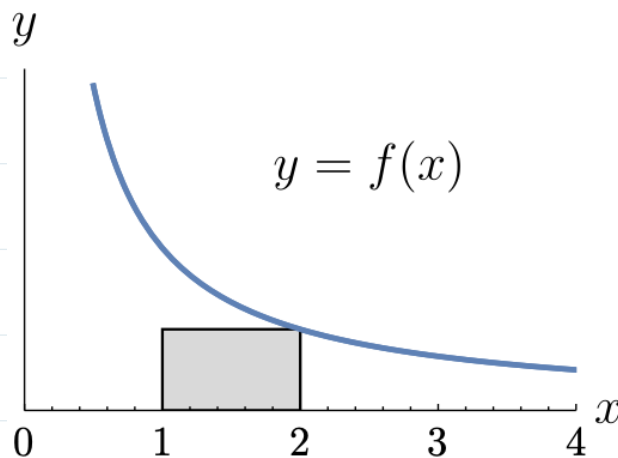
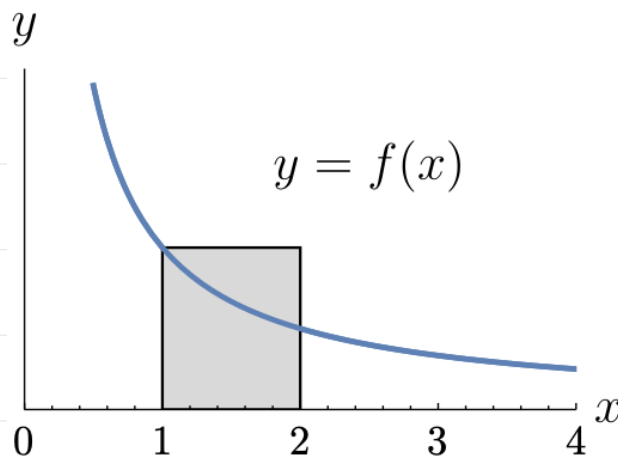
converges.

Consider f on $[1,2]$

$$a_1 = f(1), \quad a_2 = f(2)$$

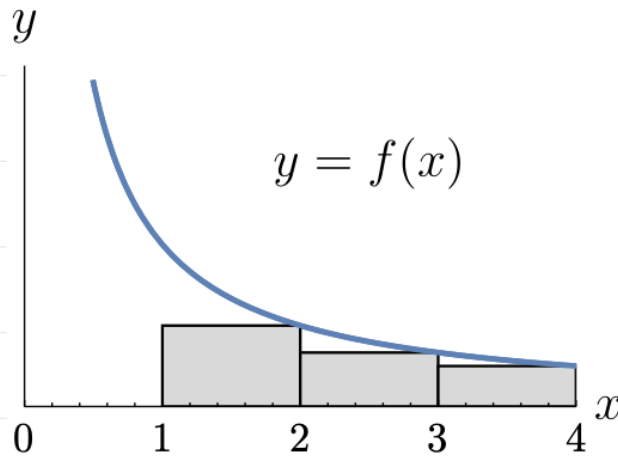
The rectangle of size $1 \times a_1$, is bigger than the rectangle of size $1 \times a_2$, and

$$a_2 < \int_1^2 f(x) dx < a_1.$$



This holds on every interval $[n, n+1]$ and we can add up all of the terms to see

$$\sum_{k=1}^{\infty} a_{k+1} = \sum_{k=2}^{\infty} a_k \leq \int_1^{\infty} f(x) dx \leq \sum_{k=1}^{\infty} a_k.$$



Now

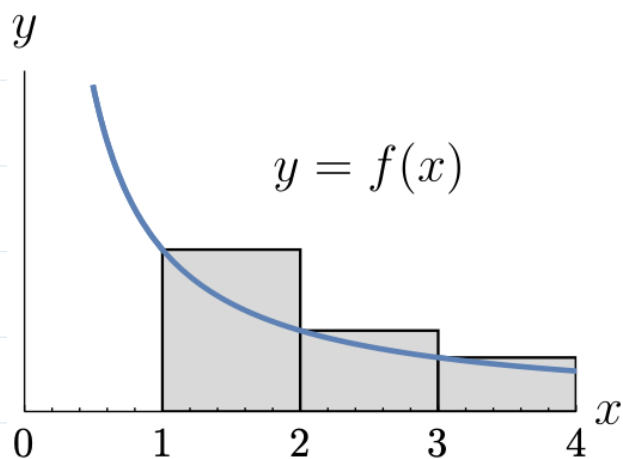
$$S_n = \sum_{k=1}^n a_k$$

is increasing since $a_k > 0$ and so if

$$\int_1^{\infty} f(x) dx$$

exists the MCT says $\sum_{k=2}^{\infty} a_k$ converges. Thus

(S_n) converges as well.



If

$$\int_1^{\infty} f(x) \, dx = \infty$$

then the larger term $\sum_{k=1}^{\infty} a_k$ diverges to ∞ .

Example #1

Determine if

$$\sum \frac{1}{n^2}$$

converges.

Let

$$f(x) = \frac{1}{x^2}$$

for $x > 0$. With x^2 positive and increasing, f is positive and decreasing. (Clearly f is continuous.)

Now

$$\begin{aligned}\int_1^{\infty} \frac{1}{x^2} dx &= \lim_{b \rightarrow \infty} \left(-\frac{1}{x} \right) \Big|_{x=1}^b \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{b} + 1 \right) = 1\end{aligned}$$

tell us

$$\sum \frac{1}{n^2}$$

converges by the Integral Test. However, we do not know the value of the sum!

Example #2

Determine if

$$\sum \frac{1}{n}$$

converges.

Let

$$f(x) = \frac{1}{x}$$

for $x > 0$. With x positive and increasing, f is positive and decreasing. (Clearly f is continuous.)

Now

$$\begin{aligned}\int_1^{\infty} \frac{1}{x} dx &= \lim_{b \rightarrow \infty} \ln|x| \Big|_{x=1}^b \\ &= \lim_{b \rightarrow \infty} \ln(b) - \ln(1) \\ &= \infty\end{aligned}$$

tell us

$$\sum \frac{1}{n}$$

diverges by the Integral Test. This happens even though

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Thus the terms in the series do not tend to zero fast enough to ensure convergence.

The rate at which the series grows is very slow:

$$\sum_{k=1}^{2^{64}} \frac{1}{k} \approx 45.$$

Definition #1 (p -series)

Let $p \in \mathbf{R}$.

The sum

$$\sum \frac{1}{n^p}$$

is a p -series.

Theorem #2

The p -series

$$\sum \frac{1}{n^p}$$

converges if and only if $p > 1$.

Definition #2 (Harmonic Series)

We call

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

the **harmonic series**.

Theorem #3 (Integral Test Error Estimate)

Assume f and (a_n) satisfy the hypotheses of the Integral Test and

$$S = \sum_{k=1}^{\infty} a_k$$

converges.

Then

$$\int_{n+1}^{\infty} f(x) \, dx < S - S_n < \int_n^{\infty} f(x) \, dx$$

where S_n is the n th partial sum of S .

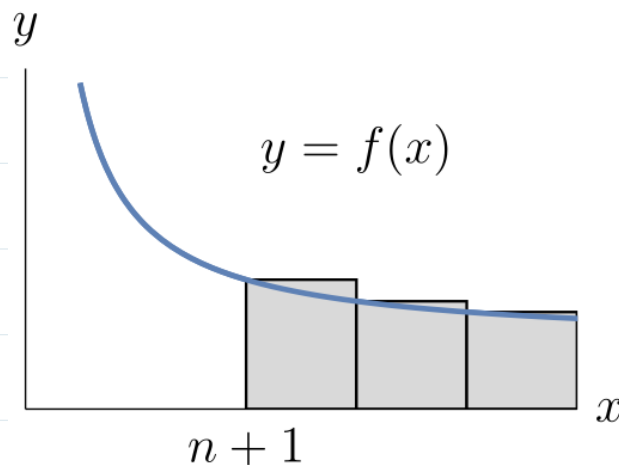
We want to approximate the tail R_n ,

$$0 < R_n = S - S_n = \sum_{k=n+1}^{\infty} a_k = a_{n+1} + a_{n+2} + \dots,$$

or how close the full sum S is to the partial sum S_n .

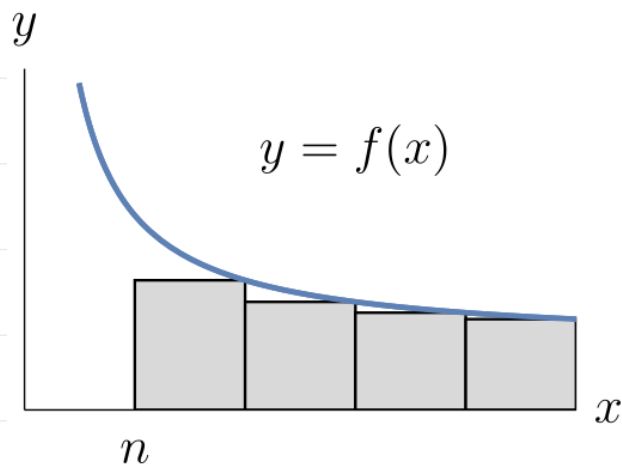
Observe

$$\int_{n+1}^{\infty} f(x) dx < \sum_{k=n+1}^{\infty} a_k$$



and shift the rectangles to the left so we can compare their area to f 's integral over $[n, \infty)$

$$\sum_{k=n+1}^{\infty} a_k < \int_n^{\infty} f(x) dx.$$



Hence

$$\int_{n+1}^{\infty} f(x) dx < \sum_{k=n+1}^{\infty} a_k < \int_n^{\infty} f(x) dx.$$

and so if we add S_n to each term we find

$$S_n + \int_{n+1}^{\infty} f(x) dx < S_n + \sum_{k=n+1}^{\infty} a_k < S_n + \int_n^{\infty} f(x) dx$$

$$S_n + \int_{n+1}^{\infty} f(x) dx < S < S_n + \int_n^{\infty} f(x) dx$$

since

$$S = S_n + R_n.$$

Remark

The following are equivalent:

1.

$$\int_{n+1}^{\infty} f(x) \, dx < S - S_n < \int_n^{\infty} f(x) \, dx$$

2.

$$S_n + \int_{n+1}^{\infty} f(x) \, dx < S < S_n + \int_n^{\infty} f(x) \, dx$$

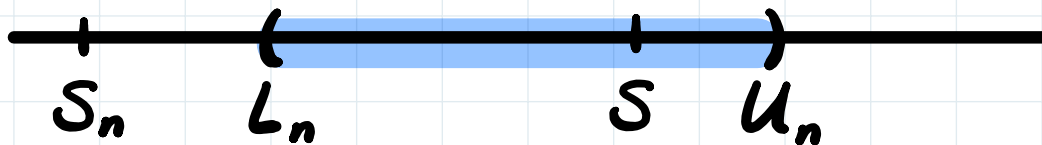
The term

$$S - S_n$$

measures the size of the error between the exact sum and the partial sum.

The terms

$L_n = S_n + \int_{n+1}^{\infty} f(x) dx$ and $U_n = S_n + \int_n^{\infty} f(x) dx$
are altogether different.



We have

$$S \in (L_n, U_n)$$

and

$$S_n < L_n$$

because $\int_{n+1}^{\infty} f(x) dx > 0$.

This means that

$$L_n$$

is a better approximation of S than S_n !

Example #3

Let

$$S = \sum_{k=1}^{\infty} \frac{1}{k^2}.$$

1. Approximate S using S_{10} and estimate the error.
2. How many terms are required to ensure that the approximation by S_n has an error less than 10^{-2} ?

We have

$$S_{10} = \sum_{k=1}^{10} \frac{1}{k^2} \approx 1.5498.$$

The Integral Test Error Estimate gives

$$\int_{11}^{\infty} \frac{1}{x^2} dx < S - S_{10} < \int_{10}^{\infty} \frac{1}{x^2} dx.$$

Since

$$\int_n^{\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} -\frac{1}{x} \Big|_{x=n}^b = \frac{1}{n},$$

we have

$$\frac{1}{11} < S - S_{10} < \frac{1}{10}.$$

Thus

$$S \approx 1.5498$$

with an error less than 0.1.

To obtain an error less than 10^{-2} , it is enough to choose n such that

$$\int_n^{\infty} \frac{1}{x^2} dx = \frac{1}{n} < \frac{1}{10^2}.$$

This occurs when

$$10^2 < n.$$

Therefore, we can choose

$$n = 10^2 + 1 = 101$$

to get

$$S - S_{101} < 10^{-2}.$$

Example #4

Let

$$S = \sum_{k=1}^{\infty} \frac{1}{k^2}.$$

Use L_{10} to obtain a better approximation of S and estimate the error.

We have

$$\begin{aligned}L_{10} &= S_{10} + \int_{11}^{\infty} \frac{1}{x^2} dx \\&= S_{10} + \frac{1}{11} \\&\approx 1.6407\end{aligned}$$

and

$$\begin{aligned}U_{10} &= S_{10} + \int_{10}^{\infty} \frac{1}{x^2} dx \\&= S_{10} + \frac{1}{10} \\&\approx 1.6498.\end{aligned}$$

Therefore,

$$S \in [1.6407, 1.6498].$$

In particular,

$$S \approx L_{10} \approx 1.6407$$

with error

$$\begin{aligned}
0 < S - L_{10} &< \underbrace{U_{10} - L_{10}}_{\text{length of } [L_{10}, U_{10}]} \\
&= \frac{1}{10} - \frac{1}{11} \\
&= \frac{1}{110} \\
&\approx 0.0091 \\
&< 10^{-2}.
\end{aligned}$$

Only using S_{10} gave an error smaller than 0.1!
 To obtain an error bound for S_n comparable to the 0.0091 bound for L_{10} , we need $n = 101$.

Remark

With some work, we can show

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$